Annabelle Zhu



Full-Stack Developer | Frontend & Python Backend

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Portfolio annabelle0726.github.io/Annabelle0726

Profile

Data Science graduate student focused on frontend and Python backend development. Independently launched Agency Bloom; contributed to Belay's learning workspace, session recovery and Docker sandbox. Apply Vibe Coding with manual code review, integration and regression checks.

Education

University of Arizona | M.S. in Data Science, in progress — 2025 - Present

University of Arizona | B.A. in Information Science & eSociety — 2022 - 2024

Selected Projects

Agency Bloom

Independent full-stack development | WeChat Mini Program | Launched Case / GitHub

- **Full-stack delivery.** Built a Vue 3, TypeScript and uni-app client with a Python / FastAPI backend, connecting onboarding, AI dialogue, decision cards and action records through integration and launch.
- **Backend and data.** Integrated WeChat login, authorization and DeepSeek; persisted users, sessions and cards with SQLAlchemy and handled model responses, request status and costs.
- **Frontend.** Implemented multi-turn dialogue, themes, card confirmation, editing, pausing and progress updates. Demonstrated the complete product through a case study and interaction demo.

Belay

AI learning tool | Collaborative full-stack development | Aug 2026 - Present Case / GitHub

- **Learning workspace.** Reorganized exercises, editor, results and AI dialogue; supported Markdown, code blocks and quoted follow-ups to connect code execution with feedback.
- **Python backend.** Contributed to FastAPI persistence, authorization and session recovery; preserved drafts and request IDs and handled rate limits and uncertain outcomes to prevent duplicate submissions.
- **Docker sandbox.** Contributed to Python execution in Docker with networking disabled, read-only filesystems and non-root users; enforced CPU, memory, process and time limits and container cleanup.
- **Boundaries and tests.** Implemented bounded output and artifact extraction; contributed to host-side grading separation. Added boundary, timeout and cleanup tests with real-container validation records.

Somnia Library

Reading management app | Extended from Reverie Case / GitHub

- **Frontend and auth.** Improved homepage, book and authentication flows connecting discovery, account access and library management. Clarified search, favorites and add-book actions.
- **Interactions.** Refined book-cover, title and card hierarchy, navigation and state feedback. Showcased homepage, library and guided selection flows through interaction recordings.

Frontend Interactions

Cupcakes / Travel Discovery Case

- **Cupcakes.** Contributed a flavor-reveal interaction linking visual cues, clicks and immediate feedback across initial and revealed states.
- **Travel Discovery.** Contributed preference selection and destination cards in a course prototype, connecting beach, city and mountain choices with browsing and reselection.

Skills

Frontend Vue 3, React, TypeScript, JavaScript, HTML/CSS, uni-app; components and API integration.

Backend Python, FastAPI, SQLAlchemy, SQL, Docker; APIs, authorization, persistence and execution isolation.

Vibe Coding ChatGPT, Claude Code, Codex; requirements breakdown, AI-assisted coding, manual review, Git and regression testing.